

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.2: Logic Gates and Logic Circuits

Concept 3.2.2: Truth Tables

Total Questions: 15

Mark Schemes begin on Page 12

Question 1 | Source: 9618_w25_qp_12 (Q9.b)

(b) Consider the following logic expression:

$$X = \text{NOT} ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

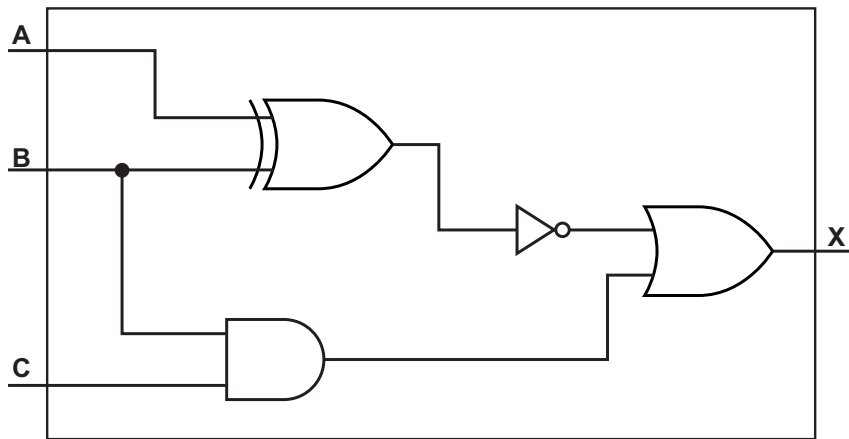
Error 2 Row number

Error 3 Row number

[2]

Question 3 | Source: 9618_s25_qp_11 (Q1.b)

(b) Complete the truth table for the following logic circuit.



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 4 | Source: 9618_s25_qp_12 (Q4.b)

(b) Complete the truth table for the logic expression:

$$X = \text{NOT} (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 5 | Source: 9618_s25_qp_13 (Q4.b)

(b) Complete the truth table for the logic expression:

$$X = (A \text{ OR } B) \text{ XOR } (B \text{ OR } C) \text{ XOR } (\text{NOT } A \text{ NAND } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 6 | Source: 9618_w24_qp_12 (Q1.a)

1 (a) Complete the truth table for the logic expression:

$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 7 | Source: 9618_w23_qp_11 (Q4.a)

- 4 (a) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ NAND } B) \text{ XOR } (\text{NOT } B \text{ AND } (B \text{ NOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 8 | Source: 9618_w23_qp_12 (Q4.b)

- (b) Complete the truth table for the logic expression:

$$X = A \text{ XOR } (B \text{ AND } (A \text{ NAND } B) \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 9 | Source: 9618_w23_qp_13 (Q4.a)

- 4 (a) Complete the truth table for the logic expression:

$$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 10 | Source: 9618_s23_qp_12 (Q6.b)

- (b) Complete the truth table for this logic expression:

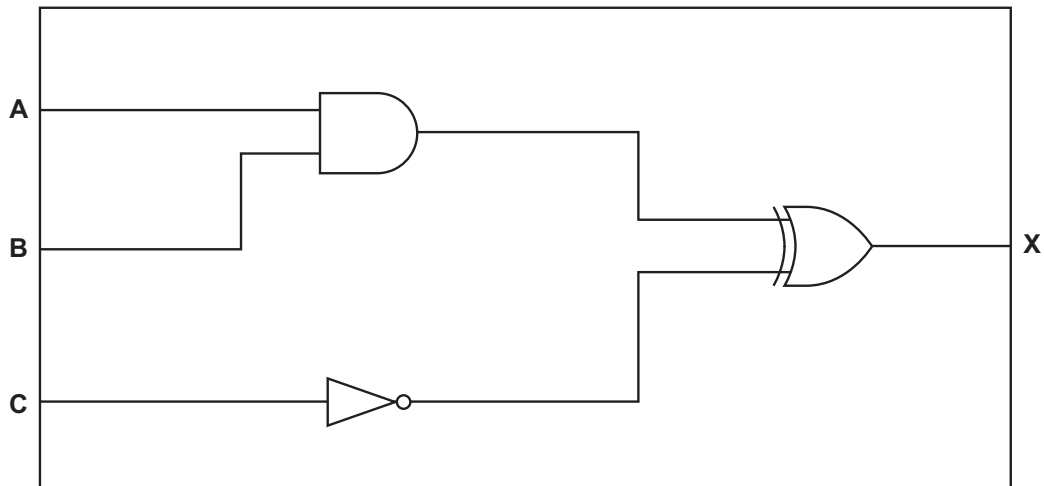
$$Z = (\text{NOT } P \text{ OR } Q) \text{ XOR } (R \text{ NOR } Q)$$

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 11 | Source: 9618_s23_qp_13 (Q1.b)

(b) Complete the truth table for this logic circuit:



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 12 | Source: 9618_w22_qp_11 (Q3.b)

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 13 | Source: 9618_w22_qp_12 (Q3.b)

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 14 | Source: 9618_w22_qp_13 (Q5.b)

(b) Complete the truth table for the logic expression:

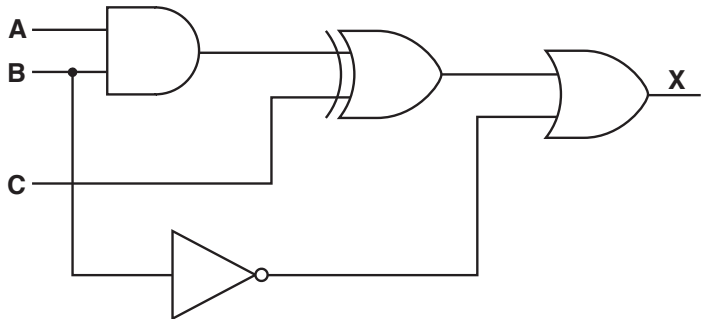
$$Y = (\text{NOT } P \text{ AND } Q) \text{ OR } (Q \text{ AND NOT } R)$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 15 | Source: 9618_w21_qp_11 (Q3.b)

3 A logic circuit is shown:



(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Question 16 | Source: 9618_s21_qp_12 (Q3.b)

3 A logic expression is given:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q9.b)

9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q3.a)

3(a)

1 mark per shaded area, max 2 marks

2

A	B	C		X
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q1.b)

1(b)

1 mark for first 4 rows, **1 mark** for second 4 rows

2

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		1

Mark Scheme for Question 4 | Source: 9618_s25_ms_12 (Q4.b)

4(b)

1 mark for first four rows, **1 mark** for second four rows

2

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		0
1	1	1		1

Mark Scheme for Question 5 | Source: 9618_s25_ms_13 (Q4.b)

4(b)

1 mark for 1st 4 rows
1 mark for 2nd 4 rows

2

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		1

Mark Scheme for Question 6 | Source: 9618_w24_ms_12 (Q1.a)

1(a)

1 mark for each shaded part.

2

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Mark Scheme for Question 7 | Source: 9618_w23_ms_11 (Q4.a)

4(a)

1 mark for each set of 4 rows (shaded)

2

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Mark Scheme for Question 8 | Source: 9618_w23_ms_12 (Q4.b)

4(b)

1 mark for each set of rows (shaded)

2

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Mark Scheme for Question 9 | Source: 9618_w23_ms_13 (Q4.a)

4(a)

1 mark for each set of highlighted rows

2

P	Q	R	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Mark Scheme for Question 10 | Source: 9618_s23_ms_12 (Q6.b)

6(b)

1 mark for first 4 rows
1 mark for last 4 rows

2

P	Q	R	Working space	Z
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		1
1	1	1		1

Mark Scheme for Question 11 | Source: 9618_s23_ms_13 (Q1.b)

1(b)

1 mark for first 4 rows correct; **1 mark** for second 4 rows correct

2

Mark Scheme for Question 12 | Source: 9618_w22_ms_11 (Q3.b)

3(b)

1 mark for each set of rows as highlighted:

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

2

Mark Scheme for Question 13 | Source: 9618_w22_ms_12 (Q3.b)

3(b)

1 mark for each set of rows as highlighted:

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

2

Mark Scheme for Question 14 | Source: 9618_w22_ms_13 (Q5.b)

5(b)

1 mark for each set of highlighted rows.

2

P	Q	R	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Mark Scheme for Question 15 | Source: 9618_w21_ms_11 (Q3.b)

3(b)	1 mark for each set of 4 rows (shaded)	2																																													
<table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>			A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											

Mark Scheme for Question 16 | Source: 9618_s21_ms_12 (Q3.b)

3(b)

1 mark for each half (shaded)

2

A	B	C	Working space	S
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		1